

Hierarchical Shell-Manifold Theory: Core Geometric Structure

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July 2026

Abstract

Hierarchical Shell-Manifold Theory (HSMT) is a speculative geometric model in which spacetime structure is organised into dimensionally nested, spatially immanent hierarchical grades labelled by an integer index N . These grades are organised by a Dirac-type Master Spectral Operator acting on a graded space; residual coupling between grades is controlled by a Gaussian overlap kernel whose width is constrained by the requirement that large-scale geometry appear statistically isotropic and homogeneous. A multifractal measure supported on the grades encodes dimensional flow and the passage of structure from deeper, quantum-like grades into an effective continuum geometry at large scales. The model is presented as a work in progress whose primary content is the hierarchical-geometric construction itself.

1 Introduction

Most approaches to the geometry of spacetime begin with a smooth manifold of fixed dimension and then quantise fields or the metric on that background, or else replace the continuum by a fundamentally discrete structure. Hierarchical Shell-Manifold Theory explores a different organising principle: a hierarchy of geometric grades that are simultaneously nested in a dimensional sense and immanent in a spatial sense.

The grades $\mathcal{M}_N$, labelled by an integer N , are not thin shells occupying distinct spatial locations. Lower grades are constitutively present within higher grades as their geometric foundation, while remaining hierarchically distinct. A single Dirac-type Master Spectral Operator organises these grades through its spectral structure. Residual mixing between grades is quantified by a Gaussian kernel in an abstract hierarchical coordinate; the width of the kernel is required to suppress residual coupling sufficiently for the large-scale geometry to recover statistical isotropy, homogeneity, and effective smoothness.

On this graded arena a multifractal measure encodes both dimensional flow—an effective dimension that may vary with grade and scale—and the passage of geometric and spectral content from deeper grades into macroscopic continuum structure. In this setting, quantum-like spectral features are regarded as natural consequences of the hierarchical geometry rather than as structures imposed upon a classical continuum.

The present paper develops the core geometric objects of the model and states their mutual relations. No claim is made that the construction constitutes a complete unification of physics or a finished theory of quantum gravity. HSMT is offered as a speculative hierarchical-geometric framework whose internal coherence and continuum limit remain under active investigation.

2 Core Geometric Objects

2.1 Hierarchical Grades

The fundamental arena is a collection of hierarchical grades labelled by an integer index $N \in \mathbb{Z}$:

$$\mathcal{M} = \bigsqcup_{N \in \mathbb{Z}} \mathcal{M}_N.$$

The grades are dimensionally nested and spatially immanent. Lower grades are constitutively present throughout higher grades as their geometric foundation, while remaining hierarchically distinct. They are not thin shells occupying separate spatial locations. No single global centre of the hierarchy is required; the radial structure is understood as locally organised.

2.2 Hierarchical Radial Coordinate

The coordinate ρ that appears in residual couplings is an abstract hierarchical (spectral) radial coordinate. It measures separation between grades in the hierarchy. It is not identified with ordinary cosmological distance or with observational distortion measures.

2.3 Gaussian Residual Coupling

Residual interaction between hierarchical grades is quantified by a Gaussian overlap kernel

$$w(\rho) = \frac{1}{\sqrt{2\pi} \sigma_0} \exp\left(-\frac{\rho^2}{2\sigma_0^2}\right).$$

The kernel assigns stronger weight to neighbouring grades and rapidly weaker weight to more distant grades. Because the Gaussian never reaches exact zero, a residual coupling is always present.

The width σ_0 is constrained by the continuum-limit requirement that residual hierarchical coupling be suppressed strongly enough for the large-scale geometry to appear statistically isotropic, homogeneous, and free of detectable hierarchical contamination. The numerical value $\sqrt{2}/4$ is retained only as a provisional candidate; it is not treated as a derived or fundamental constant at the present stage.

2.4 Master Spectral Operator

The hierarchical grades are organised by a single Master Spectral Operator $\mathcal{D}_\rho$. This operator is Dirac-type (first-order and self-adjoint). It acts on a graded space

$$\mathcal{H} = \bigoplus_N \mathcal{H}_N$$

whose summands correspond to the hierarchical grades. The spectral structure of $\mathcal{D}_\rho$ defines and arranges the grades $\mathcal{M}_N$.

The square $\mathcal{D}_\rho^2$ supplies the associated Laplacian-type operator and thereby carries second-order geometric information. Matrix elements of $\mathcal{D}_\rho$ between different grades carry a residual factor given by the Gaussian kernel; matrix elements within a single grade remain unsuppressed.

2.5 Multifractal Measure and Dimensional Flow

A multifractal measure is supported on the hierarchical grades. It encodes two related flows:

- dimensional flow — an effective dimension $d_N(\ell)$ that may depend on grade N and scale ℓ , approaching the ordinary continuum dimension at large scales;
- physical-content flow — the passage of geometric and spectral structure from deeper, more quantum-like grades into macroscopic continuum structure at higher grades.

These flows are consistent with the immanent relation between the grades. Quantum-like spectral features are regarded as natural consequences of the hierarchical geometry rather than as structures imposed upon a classical continuum.

3 Coherent Geometric Picture

The objects defined above interlock as follows.

The hierarchical grades $\mathcal{M}_N$ form the basic skeleton of the geometry. They are organised by the Master Spectral Operator $\mathcal{D}_\rho$, whose spectral projections correspond to the grade subspaces of $\mathcal{H}$. Because the grades stand in an immanent constitutive relation, they are never completely decoupled. The Gaussian kernel quantifies the residual overlap between them, with the width σ_0 required to keep that residual small enough for ordinary continuum physics to emerge at large scales.

The multifractal measure distributes geometric and spectral weight across the grades and thereby produces dimensional flow. At finer grades and smaller scales the effective dimension may differ from the continuum value; at large scales the flow approaches ordinary continuum dimension while residual hierarchical contamination is suppressed. In this setting, discrete spectral structure and residual inter-grade amplitudes arise naturally from the hierarchical geometry itself.

The resulting picture is a single hierarchical-geometric arena in which quantum-like features appear at deeper grades and an effectively smooth, isotropic, homogeneous continuum geometry is recovered at large scales. No additional classical continuum is postulated and then quantised; the continuum limit is a requirement imposed on the hierarchical structure.

4 Open Questions and Current Limitations

Several aspects of the construction remain schematic or under-determined.

1. The precise value of the Gaussian width σ_0 is not yet derived. It is constrained by the continuum-limit requirement, but a controlled argument that selects a unique numerical value is still lacking.
2. The hierarchical coordinate ρ is defined only abstractly. A more precise relation between ρ and the integer grade index N remains to be fixed.
3. The concrete realisation of the graded spaces $\mathcal{H}_N$ (the nature of the sections or spinors they contain) has not been specified.
4. The multifractal measure is characterised by its functional role; an explicit construction is still missing.
5. A quantitative demonstration that a Gaussian residual coupling of the required width indeed produces an acceptably smooth, isotropic continuum limit has not yet been given.
6. The claim that quantum-like spectral structure arises from the hierarchy is conceptual. A derivation of the full apparatus of quantum mechanics from the hierarchical geometry has not been attempted.

These open questions define the immediate agenda for further maturation of the model. They are listed here so that the present status of the construction remains transparent.

5 Conclusion

Hierarchical Shell-Manifold Theory, in the form developed here, is a speculative hierarchical-geometric model organised around dimensionally nested and spatially immanent grades, a Dirac-type Master Spectral Operator, Gaussian residual coupling, and a multifractal measure that encodes dimensional and physical-content flow. The construction yields a coherent geometric picture in which quantum-like spectral structure can arise at deeper grades while an effectively smooth continuum geometry is recovered at large scales.

The model is offered strictly as a work in progress. Its value at the present stage lies in the internal coherence of the hierarchical-geometric skeleton and in the clarity with which the remaining open questions can be stated. Further development will focus on tightening the continuum-limit constraint, making the hierarchical coordinate and graded spaces more definite, and constructing the multifractal measure explicitly.

No claim is made that the present framework constitutes a complete unification of fundamental physics. It is a hierarchical-geometric proposal whose consistency, continuum limit, and possible empirical consequences remain under investigation.

References

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